

Intro

Experience with Data Preparation Tasks and Tools

Master's thesis research – Anonymous Questionnaire

This questionnaire aims to understand participants' prior experience with data preparation tasks and the tools used to support them, as well as their expectations regarding intelligent assistance. No technical background is required.

Data preparation refers to the set of activities performed to make raw data ready for analysis or modelling, such as cleaning missing values, correcting errors, transforming formats, and selecting relevant variables.

This study is conducted by Diogo Andrade, NOVA IMS. Participation is voluntary and may be stopped at any time. All responses are anonymous and used exclusively for academic research. No personally identifiable information will be collected or disclosed. By proceeding, you confirm that:

- You have read and understood the information above.
- You are at least 18 years old.
- You voluntarily agree to participate in this study.

In case of any questions, contact:
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- ☐ I agree to participate in the study
- ☐ I do not agree to participate

Profile

Profile

What best describes your current situation?

- ☐ Undergraduate student
- ☐ Master's student
- ☐ Doctoral student
- ☐ Working professionally
- ☐ Other:

What is your primary field of study or professional area?

- ☐ Data Science / Analytics
- ☐ Computer Science / Software Engineering
- ☐ Information Systems
- ☐ Mathematics / Statistics
- ☐ Engineering (non-computing)
- ☐ Management / Business / Economics
- ☐ Social Sciences
- ☐ Health / Life Sciences
- ☐ Arts / Humanities
- ☐ Other:

To what extent is working with data part of your current academic or professional activities?

- ☐ I rarely engage in tasks involving data
- ☐ I occasionally work on tasks that involve analysing or handling data
- ☐ I regularly work with data as part of my studies or job
- ☐ Working with data is a central part of my role

General Experience with Data Preparation

General Experience with Data Preparation

1. Before this questionnaire, how familiar were you with the concept of data preparation?

Not familiar at all

☐ ☐ ☐ ☐ ☐

Very familiar

2. Approximately how many times have you performed data preparation tasks in the past?

- ☐ Never
- ☐ 1-5 times
- ☐ 6-19 times
- ☐ 20-39 times
- ☐ More than 40 times

3. Which of the following approaches have you used to support data preparation tasks?

- ☐ Spreadsheet-based tools (e.g., Excel, Power Query)
- ☐ Programming scripts or queries (e.g., Python, R, SQL)
- ☐ Dedicated data preparation tools (e.g., OpenRefine, KNIME, Trifacta)
- ☐ Data preparation systems with integrated LLMs that automatically apply transformations
- ☐ Conversational AI assistants (e.g., ChatGPT, Claude) used to ask for help (e.g., explanations, code suggestions, debugging)

- ☐ Conversational AI assistants used to suggest data cleaning or transformation steps (e.g., generating code or describing actions to perform)

4. Which type of tool do you use most often for data preparation?

- ☐ Spreadsheet-based tools (e.g., Excel, Power Query)
- ☐ Programming scripts or queries (e.g., Python, R, SQL)
- ☐ Dedicated data preparation tools (e.g., OpenRefine, KNIME, Trifacta)
- ☐ Data preparation systems with integrated LLMs that automatically apply transformations
- ☐ Other:

5. Which specific tool do you use most often within the category you just selected?

Please specify just the name of the tool (e.g., Excel, Python, OpenRefine)

6. When performing data preparation tasks, in approximately what percentage of situations do you use

conversational AI assistants to help you? (e.g., ChatGPT, Claude)

- ☐ 0% (I never use them)
- ☐ 1-25% of the time
- ☐ 26-50% of the time
- ☐ 51-75% of the time
- ☐ 76-100% of the time

7. Which conversational AI assistant do you most frequently use to ask for help with data preparation tasks?

- ☐ ChatGPT
- ☐ Claude
- ☐ Gemini / Bard
- ☐ Perplexity
- ☐ Copilot (e.g., GitHub Copilot)
- ☐ I do not use conversational AI assistants for these tasks
- ☐ Other:

Perceived difficulty of tasks

Perceived difficulty of tasks

8. Below you will find a list of common data preparation tasks. For each task, please indicate how difficult you personally find it to perform, using a scale from Very Easy to Very Difficult. Your answers should reflect your own experience, regardless of the tools you typically use.

	Very Easy	Easy	Neither easy nor difficult	Difficult	Very Difficult
8.1 Filling missing values	☐	☐	☐	☐	☐
8.2 Detecting and handling errors and inconsistencies in raw data	☐	☐	☐	☐	☐
8.3 Identifying and handling outliers/extreme values	☐	☐	☐	☐	☐
8.4 Understanding the semantic meaning of columns	☐	☐	☐	☐	☐
8.5 Deciding how to transform or encode data for analysis	☐	☐	☐	☐	☐
8.6 Deciding which data and features should be included or excluded before modelling	☐	☐	☐	☐	☐

Familiarity with LLMs applied to Data Preparation

Familiarity with LLMs applied to Data Preparation

9. Before this study, were you aware that AI/LLM-based systems can be used not only to suggest data preparation steps, but also to help apply them within a data preparation tool?

- ☐ Yes, and I have used such systems
- ☐ Yes, but I have never used such systems
- ☐ No, I was not aware of this possibility

10. How confident do you feel about using data preparation systems with integrated LLMs that automatically apply transformations?

Not confident at all

☐ ☐ ☐ ☐ ☐

Very confident

Expectations about automation and control

11. Below you will find a set of statements regarding data preparation systems. For each statement, please indicate the extent to which you agree, using a scale from "Strongly disagree" to "Strongly agree".

	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
11.1 Systems that automatically suggest or apply data preparation steps can significantly reduce the effort required for data preparation	☐	☐	☐	☐	☐
11.2 I prefer to retain control over data transformations rather than fully automating them	☐	☐	☐	☐	☐
11.3 I trust automated data preparation systems more when I can inspect or modify the steps they perform	☐	☐	☐	☐	☐
11.4 Transparency and reproducibility are important when using intelligent data preparation tools	☐	☐	☐	☐	☐

Role of Intelligent Systems

Role of Intelligent Systems

12. In your opinion, what should be the ideal role of intelligent systems in data preparation?

- ☐ Fully automate the process without user intervention
- ☐ Automatically perform tasks, while allowing user inspection and control
- ☐ Assist mainly in complex or ambiguous tasks, leaving execution to the user
- ☐ I am unsure

13. What feature would you consider essential or truly transformative in a system that automatically prepares data using large language models?